

NCCL Staging Overhead vs. Direct NVLink Loads and Stores

Python

```
def ulysses_allgather_reference():
    x = x.contiguous()
    x_size = torch.tensor(x.size())...
    size_list = [torch.zeros(x_size.size()) for ...]
    dist.all_gather(size_list, x_size)
    tensor_list = [torch.empty(torch.Size...)...
    dist.all_gather(tensor_list, x, group=group)
```

C++

```
template<typename T>
__global__ void ulysses_allgather_kernel(...):
    int64_t src_idx = ...
    int64_t dst_idx = ...
    out[dst_idx] = src[src_idx];

ulysses_allgather_kernel<<<blocks, threads, 0>>>
```

Using multiple streams to pipeline data chunks

Python

```
def vocab_parallel_logprob_topk_ref():
    ...
    logits_2d = vocab_parallel_logits.reshape(...)
    target_local = target_flat[ranks * local_tokens
: (ranks + 1) * local_tokens]
    all_to_all_vp_to_seq(logits_2d, tp_group)
```

Python

```
def vocab_parallel_logprob_topk_cuda():
    ...
    for i in range(num_chunks):
        with torch.cuda.stream(streams[i]):
            streams[i].wait_stream(default_stream)
            start = i * chunk_size
            end = start + chunk_size
```